INCH-POUND

MIL-DTL-17587/1(SH)
28 June 2007

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, ACB, LOW VOLTAGE, ELECTRIC POWER, AIR, REMOVABLE CONSTRUCTION ACB-900RC, ACB-902R, AND ACB-904LRC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17587.

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Requirements:

TABLE I. Ratings/features of circuit breakers.

Circuit breaker type	**ACB-900RC**	**ACB-902R**	**ACB-904LRC**
Rated continuous current (amperes)	900	900	900
Rated maximum voltage (AC volts) /1	500	500	500
Frequency (hertz)	60	60	60
Maximum interrupting rating current (symmetrical amperes)	42,000	50,000	85,000
Rated short-time current /2 (amperes)	25,000	25,000	25,000
Rated short-time current duration (seconds)	0.5	0.5	0.5
Operating mechanism - manual	STD	STD	STD
Operating mechanism - electric	STD	STD	STD
Maximum weight (lbs) – circuit breaker (removable component)	218	122	123
Maximum weight (lbs) – cradle (stationary component)	112	26	26
Number of poles	3	3	3
Number of overcurrent coils or solid-state sensors	2/3	2	3
Attachments			
Electric closing mechanism	STD	STD	STD
Auxiliary switches (number of contacts)	8	10	10
Undervoltage lockout device	Optional for all circuit breaker types		
Mechanical position indicator	Required for all circuit breaker types		
Secondary disconnecting devices	Required for all circuit breaker types, unless otherwise specified		
Cell switch	Optional for all circuit breaker types		
Indicator light	Provisions for mounting required for all circuit breaker types; Furnished only when specified		
Hold-closed mechanism	OPT	OPT	OPT
Undervoltage release device	Optional for all circuit breaker types		
Shunt trip device	Required for electrically-operated breakers		
Operations counter	OPT	OPT	STD
Trip on power up (TOPU) device	N/A	OPT	OPT
Rack-in/rack-out position indicator	STD	STD	STD

NOTES:

/1 Voltage ratings apply to main current-carrying parts. Operating attachments shall be of voltage specified in 6.2.

/2 Ampere values apply to circuit breakers with overcurrent coils or sensors of maximum current rating within the circuit breaker type.

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)			Short-time pickup setting (amperes)						Maximum instantaneous trip /2 (amperes)
ACB-900RC	320	500	640	840	---	640	840	960	1280	1600	16000
	400	640	840	960	720	800	960	1280	1800	2100	20000
	480	720	960	1280	---	960	1280	1600	2100	2800	20000
	560	840	1120	1400	---	1120	1400	1800	2400	3200	20000
	640	960	1280	1600	---	1280	1600	2100	2800	3600	20000
	800	1280	1600	2100	1200	1600	2100	2400	3200	4000	35000
	900	1320	1800	2340	---	1800	2500	2700	3600	4800	35000
ACB-902R	100	160	200	250	---	200	250	320	400	500	2000
	160	250	320	400	240	320	400	500	720	840	2300
	250	400	500	640	570	500	640	840	1120	1400	5000
	320	500	640	840	---	640	840	960	1280	1600	6400
	400	640	840	960	---	800	960	1280	1800	2100	8000
	480	720	960	1280	720	960	1280	1600	2100	2800	9600
	560	840	1120	1400	---	1120	1400	1800	2400	3200	11200
	640	960	1280	1600	---	1280	1600	2100	2800	3600	12800
	800	1280	1600	2100	1200	1600	2100	2400	3200	4000	16000
	900	1320	1800	2340	---	1800	2500	2700	3600	4800	18000
ACB-904LRC	150	105 150	120 165	135	150 600	225 750	300 900	375 1050	450 1200	525	3000
	275	192.5 275	220 302.5	247.5	275 1100	412.5 1375	550 1650	687.5 1925	825 2200	962.5	5500
	500	350 500	400 550	450	500 2000	750 2500	1000 3000	1250 3500	1500 4000	1750	10000
	900	630 900	720 720	810	900 3600	1350 4500	1800 5400	2250 6300	2700 7200	3150	18000

NOTES:

/1 Continuous current rating.

/2 The instantaneous trip device shall be set at the place of manufacture from five times the coil or sensor rating to the maximum pickup setting specified for mechanical coils and 20 times sensor rating for solid state sensors with a default at maximum listed rating.

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TABLE III. Electrical endurance and temperature rise test requirements.

Electrical endurance test:

Type	Total number of close-open cycles	Amperes on circuit breaker close	Amperes on circuit breaker open	Maximum AC power factor (lagging) X/R
ACB-900RC	10,000	6,000	900	Close: 0.25 maximum; Open: 0.80 maximum
ACB-902R	25,000	5,400		Close: 0.25 maximum; Open: 0.80 maximum
ACB-904LRC	25,000	5,400		Close: 0.25 maximum; Open: .080 maximum

Temperature rise test:

Type	Continuous current rating	Cable quantity per terminal	Cable size /1	Bus bar quantity per phase	Bus bar size (inches) /2
ACB-900RC	900	3	300 MCM	1	2.5 x ¼
ACB-902R		3 /3		2	1.5 x ¼
ACB-904LRC		3			

NOTES:

/1 Information regarding cable is contained in MIL-DTL-915. If 3-phase cable is used for this test, the cable jacket shall be stripped for a minimum of 2 feet.

/2 Information regarding bus bars is contained in MIL-DTL-16036.

/3 Connect with bus adapter of minimum length possible.

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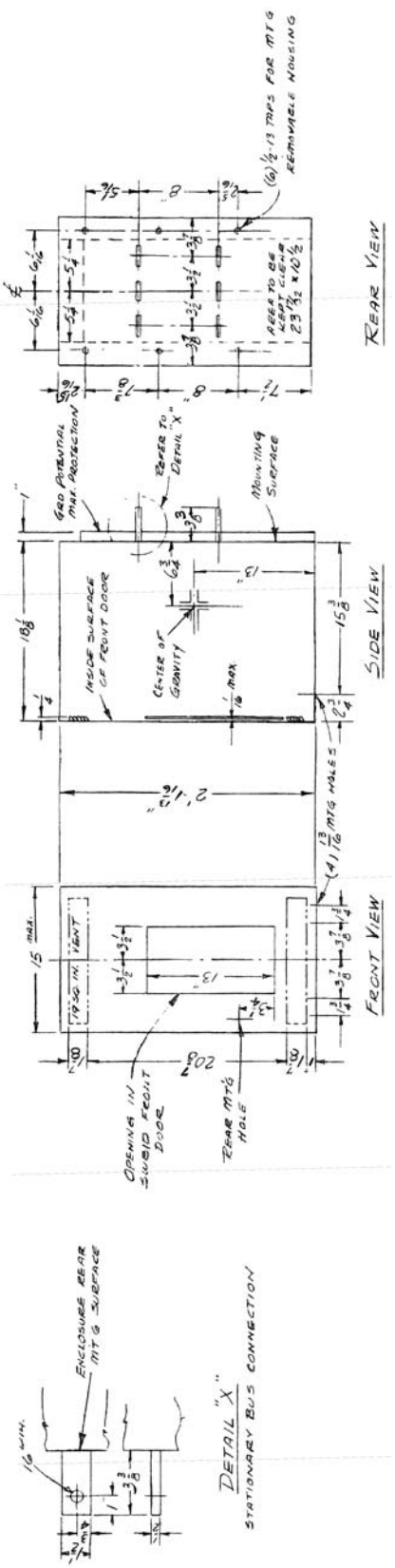


FIGURE 1. Outline and mounting dimensions for circuit breaker type ACB-900RC.

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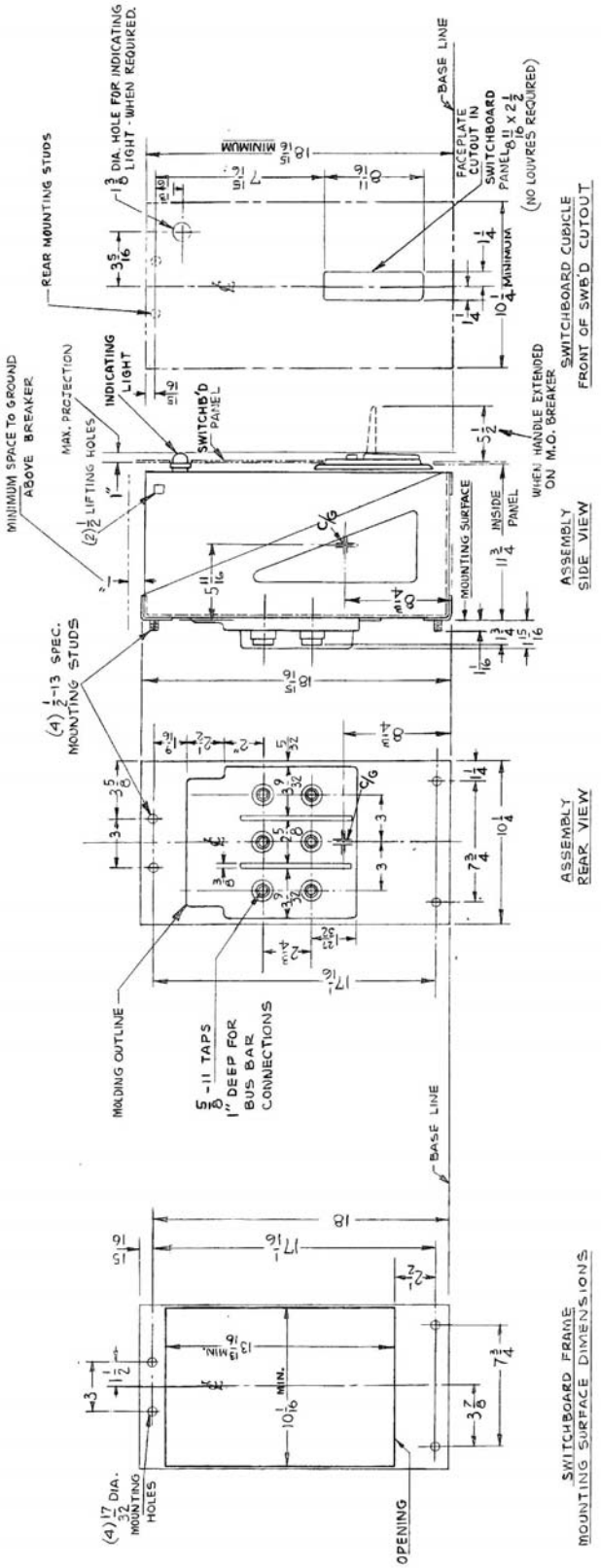


FIGURE 2. Outline and mounting dimensions for circuit breaker type ACB-902R.

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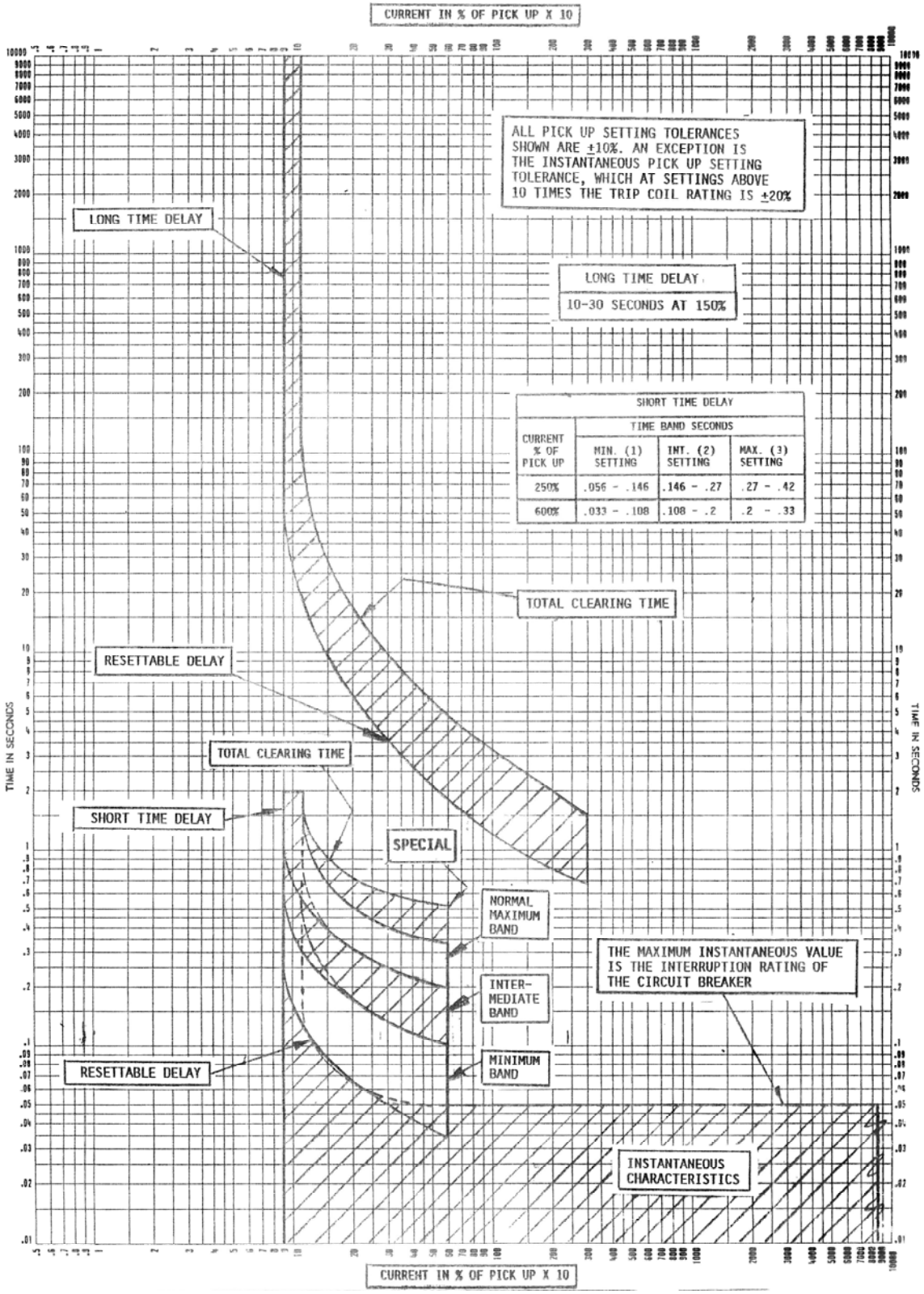


FIGURE 3. Time-current characteristic curve for circuit breaker types ACB-900RC and 902R.

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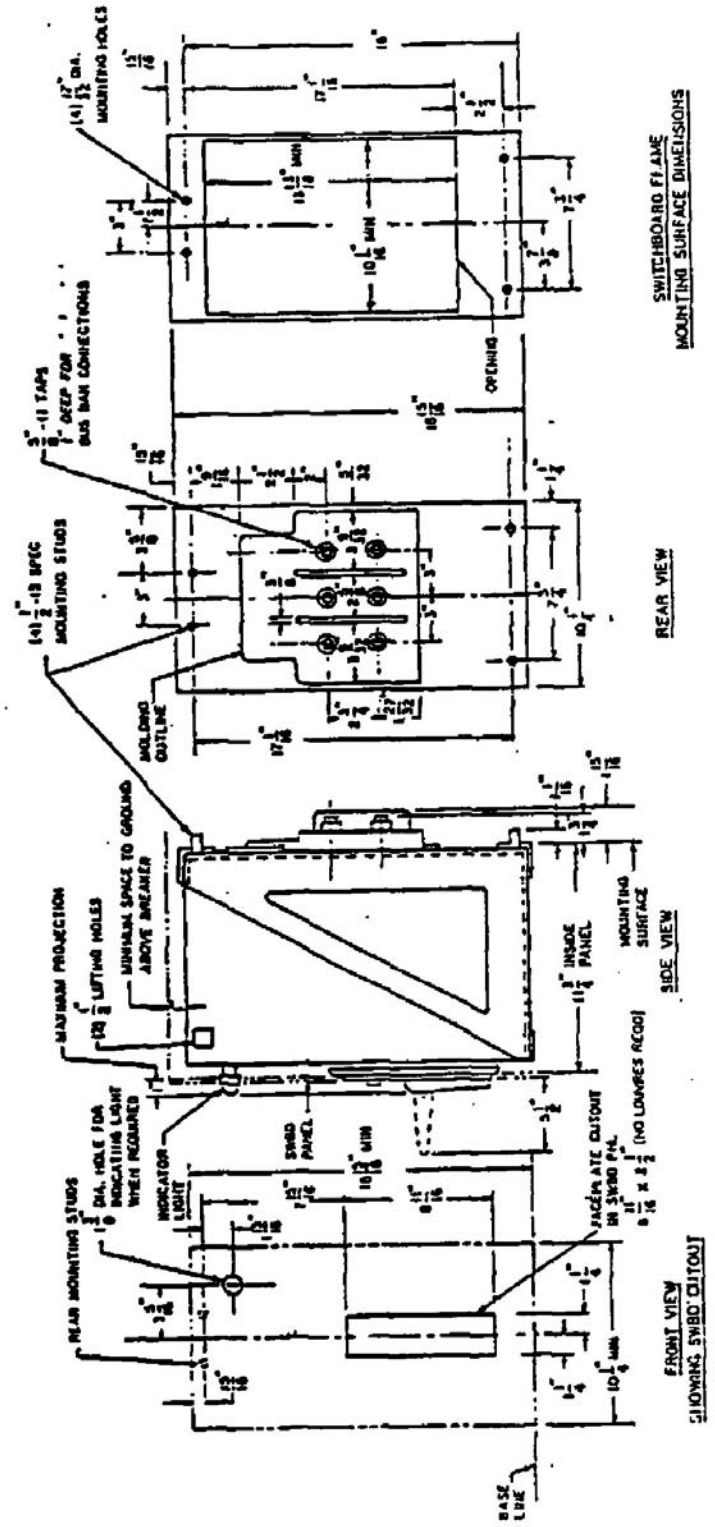


FIGURE 4. Outline and mounting dimensions for circuit breaker type ACB-904LRC.

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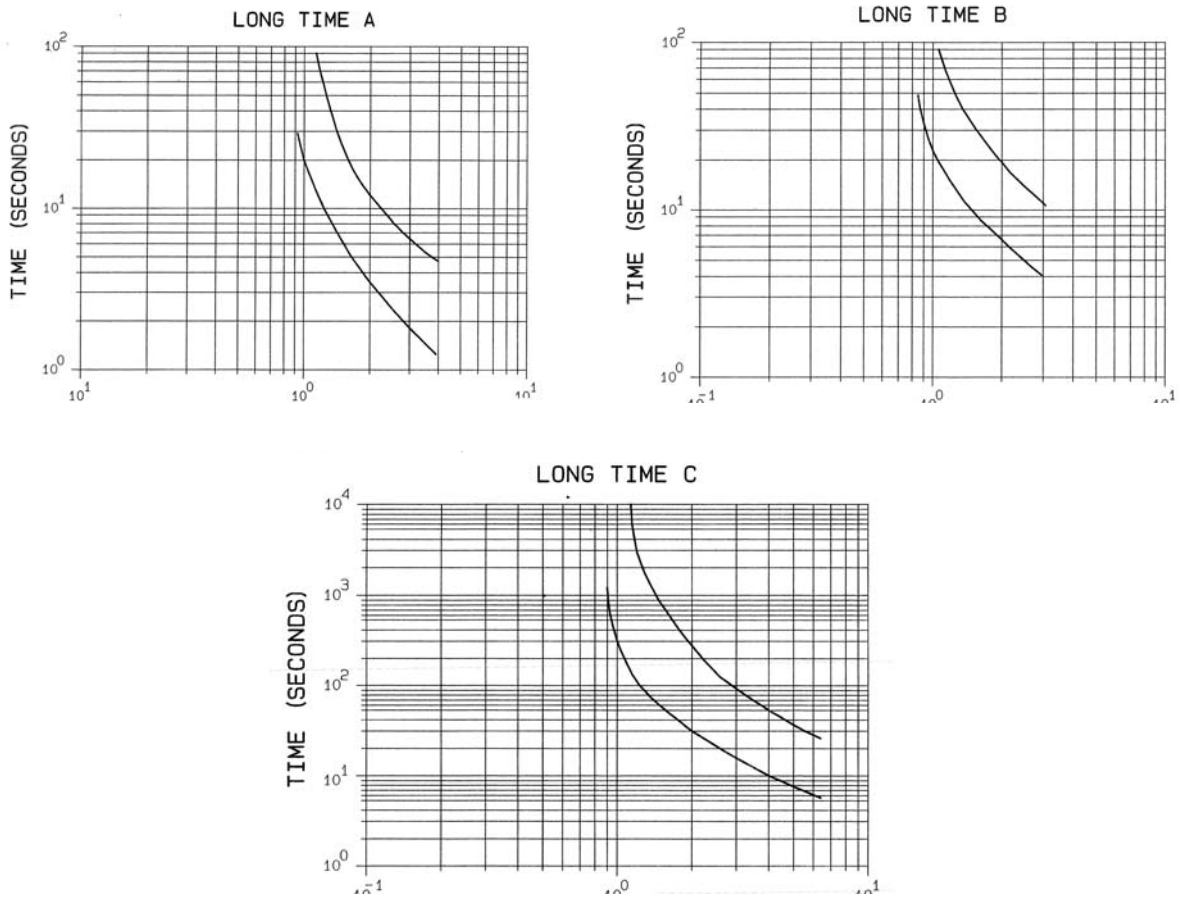


FIGURE 5. Time-current characteristic curve for circuit breaker type ACB-904LRC.

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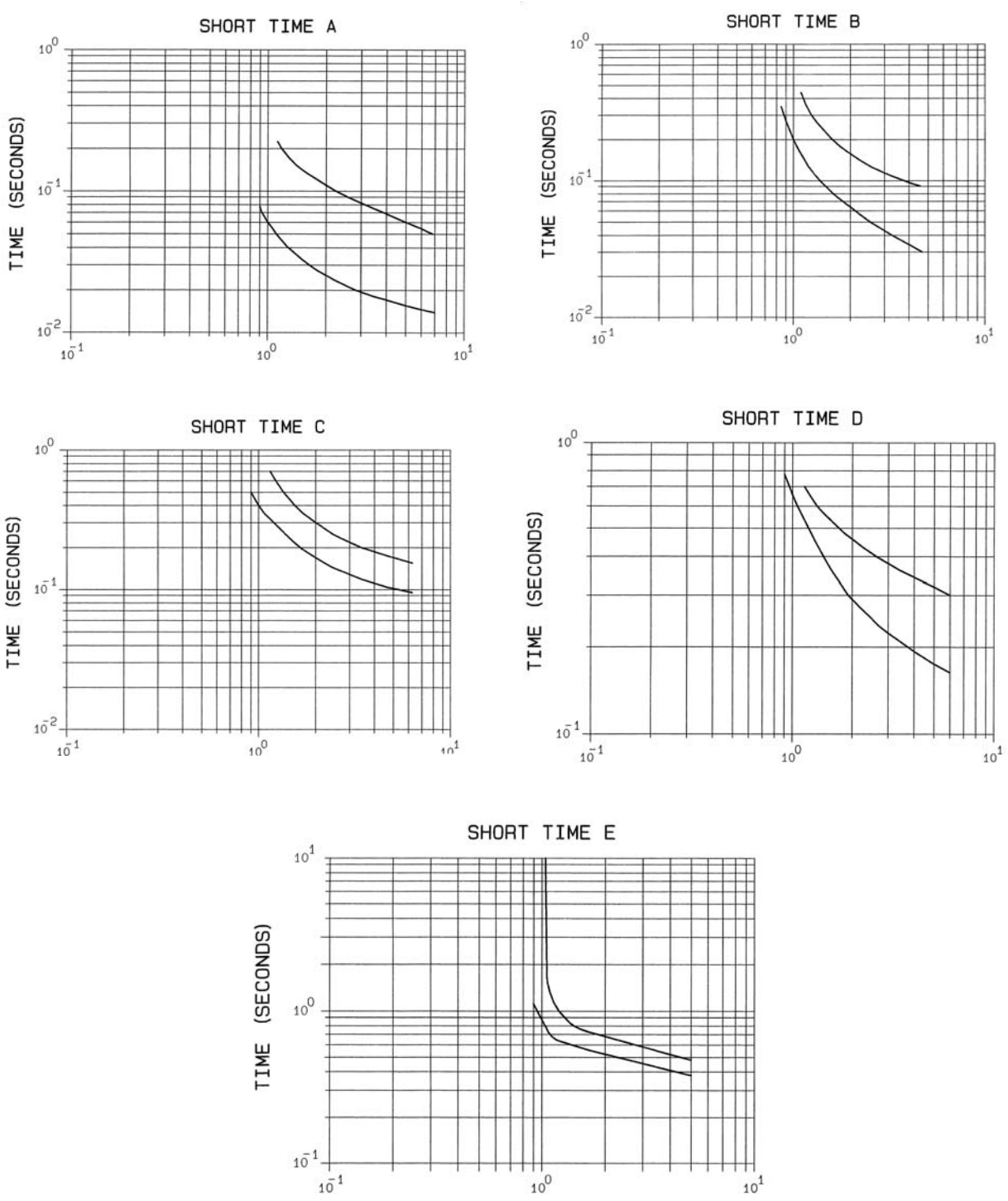


FIGURE 5. Time-current characteristics for circuit breaker type ACB-904LRC - Continued.

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